Lesson 3.2 - Mean, Median, Mode, and Range

> **Video:** Mean, Median, Mode, and Range - How To Find It! by The Organic Chemistry Tutor
> **Chapter:** 3 - Central Tendency

Definitions

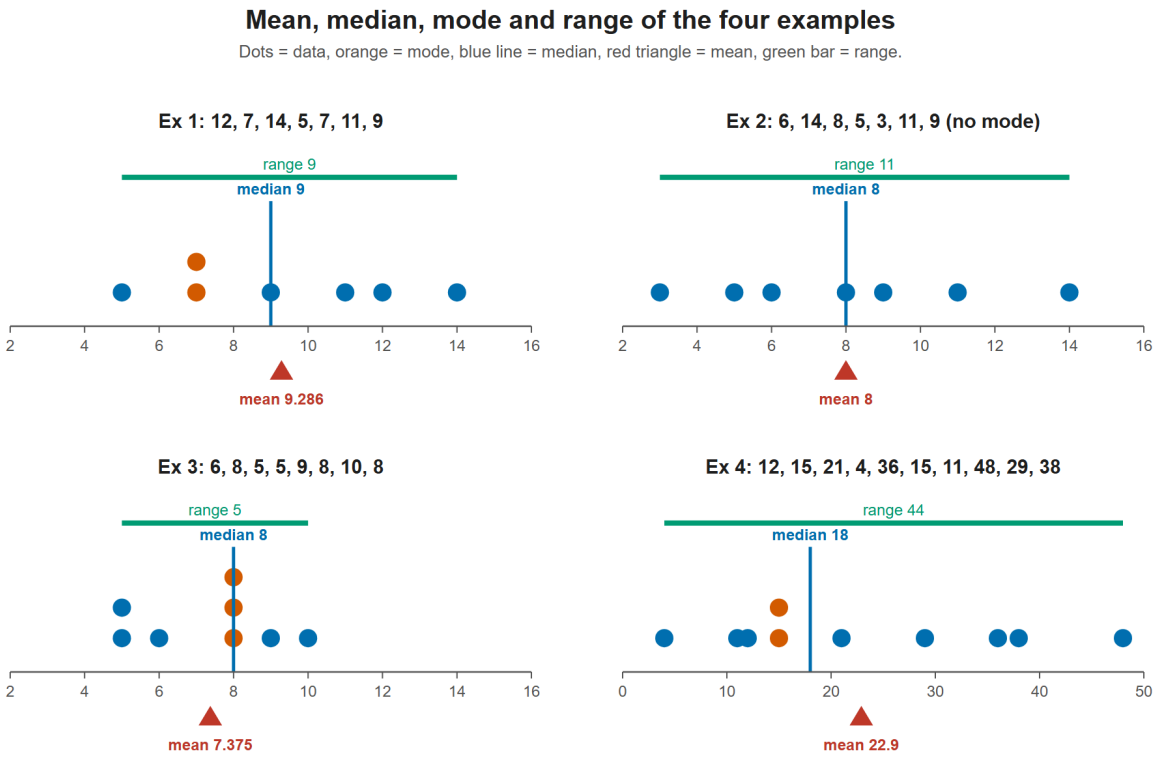
Measure	Rule
Mean (arithmetic mean, average)	Sum of the values ÷ number of values
Median	Middle value of the **sorted** data. Even n: average of the two middle values
Mode	Value that appears most often. It can have **no mode** or **several modes**
Range	Highest − lowest

Finding the median by hand: sort the data, then cross off the first and last values, then the next pair, and so on, until one value (odd n) or two values (even n) are left.

Example 1 - odd n (7 values)

Data: 12, 7, 14, 5, 7, 11, 9 → sorted: 5, 7, 7, **9**, 11, 12, 14

- Mean: $\frac{65}{7} \approx 9.29$
- Median: **9**
- Mode: **7** (the only value that appears twice)
- Range: $14 - 5 = 9$



All four examples on number lines. In Example 2 every value appears once, so there is no mode. In Example 4 the large values (36, 38, 48) pull the mean (22.9) well to the right of the median (18).

Example 2 - odd n, no mode

Data: 6, 14, 8, 5, 3, 11, 9 → sorted: 3, 5, 6, **8**, 9, 11, 14

- Mean: $\frac{56}{7} = 8$
- Median: **8**
- Mode: **none**, since every value appears once
- Range: $14 - 3 = 11$

Example 3 - even n (8 values)

Data: 6, 8, 5, 5, 9, 8, 10, 8 → sorted: 5, 5, 6, **8, 8**, 8, 9, 10

- Mean: $\frac{59}{8} = 7.375$
- Mode: **8** (appears three times)
- Median: the two middle values are both 8 → $\frac{8+8}{2} = 8$
- Range: $10 - 5 = 5$

Example 4 - even n (10 values)

Data: 12, 15, 21, 4, 36, 15, 11, 48, 29, 38 Sorted: 4, 11, 12, 15, **15, 21**, 29, 36, 38, 48

- Mean: $\frac{229}{10} = 22.9$
- Mode: **15**
- Median: $\frac{15 + 21}{2} = 18$, the midpoint of the two middle values
- Range: $48 - 4 = 44$

Key takeaways

- **Sort first.** It makes the median, mode and range easy.
- Odd n: the median is a data value. Even n: the median is the **average of the two middle values**, which may not be in the data.
- A data set can have **no mode** (all values different) or **more than one** mode.